

Heme iron intake and risk of new-onset diabetes in a Mediterranean population at high risk of cardiovascular disease: an observational cohort analysis

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Background Several epidemiological studies have observed an increased risk of type 2 diabetes mellitus (T2DM) among subjects with a higher consumption of red and processed meat. Heme iron intake has been directly associated with a higher risk of T2DM in healthy adult Chinese and U.S populations. The objective of the present study was to evaluate the association between heme iron intake and the incidence of T2DM in a Mediterranean population at high cardiovascular risk. **Methods** We assessed a subset of participants in the PREDIMED trial as an observational cohort, followed up for a maximum of eight years. We initially included 1073 non-diabetic subjects (57.1% women) aged 67.3 ± 6.0 years, at high cardiovascular risk. Diet was assessed at the study baseline using a validated, semi-quantitative food frequency questionnaire. **Results** During the follow-up period 131 diabetics were newly diagnosed. The risk of developing T2DM was assessed using baseline heme iron intake and proportional hazard models, first unadjusted, then adjusted for energy, and finally adjusted for dietary, anthropometric, socio-demographic and lifestyle variables. Significant direct associations with the incidence of T2DM were found for heme iron (Hazard Ratio [HR] 1.30, 95% confidence interval [CI], 1.02 to 1.66). Secondly, we have also observed that coffee (HR:0.93, 95% CI, 0.89 to 0.98) and alcoholic beverages (HR: 1.02, 95% CI, 1.01 to 1.04) were also found to reduce and increase the risk of T2DM, respectively. **Conclusion** High dietary intake of heme iron was associated with an increased risk of developing T2DM in a Mediterranean population at high cardiovascular risk. Trial registration Identifier: ISRCTN35739639.